

Movement Limitations: Capital

Schelling's segregation model offers a possible operationalization of mild preferences leading to large scale clustering, even when the underlying preferences driving it aren't immediately obvious to these phenomena. However, it feels distant from a residential segregation model as agent's are able to freely move without constraints. Considering the historical drivers of segregation and the connection to income, the present seeks to modify this model to incorporate capital dynamics by adding capital stock to each agent, as well as districts within the space that have their own pricing dynamics in response to its constituents.

The hypothesis is that adding such moving constraints based on wealth will produce different distributions of segregations, as well as inferences on happiness and capital for 2 different classes of agents. Although agents are still only motivated by their happiness with respect to homophily, their ability to move depends on their available capital. This could lead to agents being trapped in an unhappy situation where they may be the only ones of their group in a given cluster.

This was first initialized by providing agents with heterogeneous initial wealth, which can be adjusted in terms of its average and variability. This can then reflect social categories as well as differences in economic opportunity structures. For simplicity however, the wealth of each agent is not increasing, only staying the same or decreasing, which does represent a limitation in terms of agents stuck within groups that are likely wealthier.

This was then built into an affordability function for each agent determining their ability to move into a given random cell while still evaluating their desire to move based on homophily. If unhappy, an agent will evaluate their capital and if able to, will move to an empty plot. This is

determined by whether the slot is less than or equal to their current wealth. And naturally, this resource is finite such that if an agent is unable to find a satisfying cell before their capital runs out, they shall remain stuck. This can imply a happiness cost to agents who are disadvantaged, although such term depends on their relative distribution within the group, and between the groups.

Lastly, the district-based housing market was introduced such that the grid is partitioned into an adjustable number of grids. This implies that more grids would stimulate greater dynamics as for the same number of agents, the average wealth and thus price would be diffused, plausibly allowing for more capital to satisfy the agents' happiness. Prices only increase gradually with respect to average wealth so that changing the scale of the distribution doesn't mechanically result in the prices immediately matching this parameter and thereby rendering it futile. This can be parallel to gentrification or capitalization of districts in which wealthier residents remain happy and able to move (albeit not needing to) whereas the less wealthier and out-group individuals remain stuck (and would otherwise be driven out of the district. This reflects a path dependence in the dynamics of segregation.

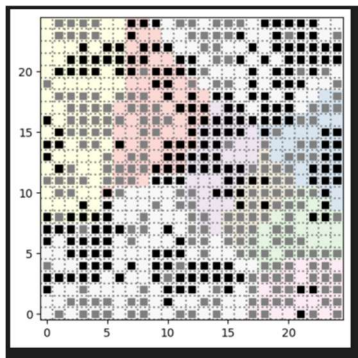
Since there's many parameters, results can be quite heterogeneous. For instance, at a high agent density, and larger number of districts, the happiness levels average around 60-70% but the amount of wealth required by each group increases, as well as the share of individuals who end up being unable to move. If wealth is uneven, e.g one group is less variable and higher capital, their remaining wealth is as expected higher, although happiness levels don't increase with this increasing expenses. If instead, there's a much lower density, happiness levels can rise overall, but the gap between the groups widens. At the minimal wealth multiplier, not much movement can happen, which means that wealth is both quickly depleted and yields a rather unhappy world.

Both surprising and also somewhat expected is that as homophily decreases, segregation actually remains pretty comparable to higher levels, perhaps a bit higher. This makes sense given the limited space and capital constraints, but consequences are that of course, more people remain satisfied whereas higher homophily can give comparable levels of segregation albeit much less happy population when keeping the district number constant.

Overall, this modification seems to have the most implications to the levels of happiness for individuals whilst maintaining pretty even levels of segregation, although this can occur as clusters within districts rather than resulting in complete segregation between districts. This is expected since agents don't end up moving as the district prices increase as their affordability, albeit can be determined if surrounding cells are smaller than the current, don't continue as though paying rent.

Example Runs:

1. High density, group 0 being less variable and higher in wealth while group 2 is more variable and less wealth. Homophily at 0.6 and max number of districts (12)



Global happiness: 90.24%

Group 0 happiness: 94.15%

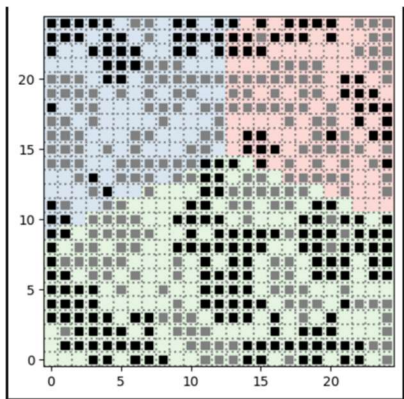
Group 1 happiness: 86.34%

Group 0 Relative Remaining Wealth: 71.28%

Group 1 Relative Remaining Wealth: 59.58%

Agents Unable To Afford Any Move: 22.20%

2. Same as above except smallest number of districts (3)



Global happiness: 94.39%

Group 0 happiness: 96.59%

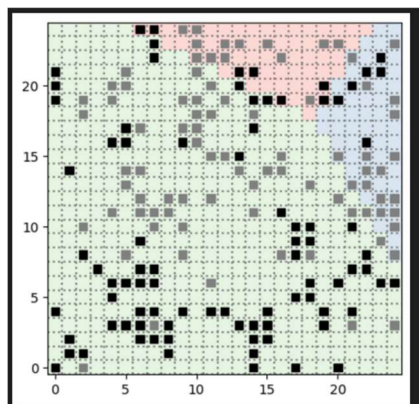
Group 1 happiness: 92.20%

Group 0 Relative Remaining Wealth: 81.15%

Group 1 Relative Remaining Wealth: 67.31%

Agents Unable To Afford Any Move: 14.63%

3. Same as (2) but changing density instead



Global happiness: 80.59%

Group 0 happiness: 85.88%

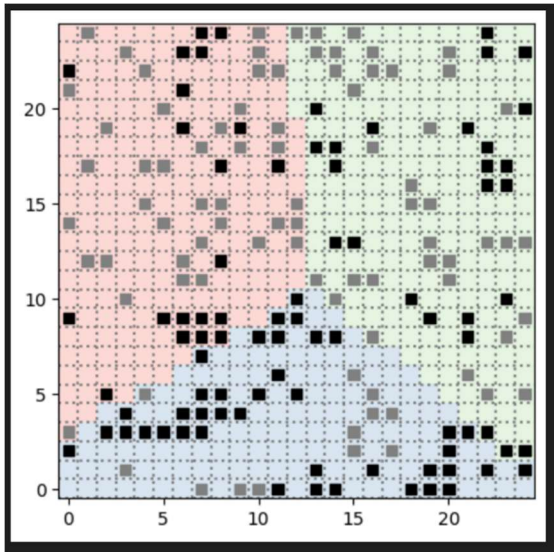
Group 1 happiness: 75.29%

Group 0 Relative Remaining Wealth: 61.21%

Group 1 Relative Remaining Wealth: 46.24%

Agents Unable To Afford Any Move: 35.29%

4. Same as (3) but more homophily (0.8)



Global happiness: 74.12%

Group 0 happiness: 80.00%

Group 1 happiness: 68.24%

Group 0 Relative Remaining Wealth: 46.29%

Group 1 Relative Remaining Wealth: 29.22%

Agents Unable To Afford Any Move: 48.24%